

Lookout 2: Anomaly Detection via Leave-One-Out Kernel Density Estimation

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Abstract

The lookout algorithm identifies anomalies in multivariate data by fitting a generalized Pareto distribution to the largest surprisals (negative log densities) of a leave-one-out kernel density estimate. Its bandwidth is chosen by persistent homology, using death diameters equivalent to the edge lengths of the Euclidean minimum spanning tree. We use the tree to analyse the algorithm and propose three changes. First, a robust covariance standardization replaces min-max scaling. Second, the bandwidth is set using an upper quantile of the edge lengths in place of the edge length below the largest gap between consecutive lengths. Third, the generalized Pareto shape parameter is constrained to be non-positive. Using limit theory for random geometric graphs, we show that the scaled quantile converges in probability to a constant for any density, so the bandwidth has the order of the spacing between neighbouring observations in the sparse part of the sample. The longest edges, which the original bandwidth typically uses, grow without bound under heavy tails. Neither bandwidth gives a consistent density estimate in general; we argue that the modified one behaves appropriately for anomaly detection. In simulations and two applications, the modified algorithm usually improves on the original and compares well with other methods.

Keywords: bandwidth selection, extreme value theory, generalized Pareto distribution, minimum spanning tree, outlier detection.

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1 Introduction

Anomaly detection is used in many application domains such as credit card fraud and scam detection (Kolupuri et al. 2025). Kandanaarachchi & Hyndman (2022) proposed the *lookout* algorithm for anomaly detection using a kernel density estimate with bandwidth based on persistent homology. Anomalies are identified by fitting a generalized Pareto distribution (GPD) to the negative logarithm of the smallest leave-one-out kernel density estimates, and declaring points with low GPD probabilities as anomalies.

In this paper we study the bandwidth of the lookout algorithm and what it implies for the kernel density estimates. The bandwidth was originally computed from the death diameters of a Vietoris-Rips filtration. These death diameters are exactly the edge lengths of the Euclidean minimum spanning tree of the data, so they can be computed more quickly from the tree, and the limit theory for minimum spanning trees applies to them. We work with the tree throughout this paper. The original algorithm sets the bandwidth using the edge length at the lower end of the largest gap between consecutive edge lengths. We propose a modified algorithm that sets the bandwidth using an upper quantile of the same edge lengths, and we show that this quantile has order $n^{-1/m}$ for any density, while under heavy tails the largest gap typically lies among edges whose lengths grow without bound. The modified algorithm is more robust than the original and improves on its performance in most cases.

1.1 Anomalies and surprisals

There is no broadly accepted definition for an anomaly. Hawkins (1980) defines an anomaly or outlier as *an observation which deviates so much from other observations as to arouse suspicion it was generated by a different mechanism*. This definition implies that anomalies are rare and different from other observations. In practice, this means fitting a probability distribution to the observations and asking whether each observation is a likely draw from it. Hyndman & Frazier (2026) proposed a formalization of this concept, which we state in the following definition.

Throughout, f is a probability density function on $\mathbb{R}^m$ with respect to Lebesgue measure. It is the density against which the observations are judged. We do not assume that every observation is a draw from f ; the anomalies are the observations that are unlikely under f .

Definition 1 (Anomalies). Given a set of observations $\mathbf{y}_i \in \mathbb{R}^m$ for $i \in \{1, \dots, n\}$, and a probability density function f , we define observation $\mathbf{y}_i$ as anomalous if

$$\mathbb{P}(f(\mathbf{Y}) \leq f(\mathbf{y}_i)) < \alpha, \quad (1)$$

where $\mathbf{Y}$ is a random vector with density f , and $\alpha > 0$ is a chosen threshold.

Definition 2 (Surprisals). Given a probability density function f , the *surprisal* of an observation $\mathbf{y}$ with respect to f is defined as $-\log f(\mathbf{y})$.

These are better known as “log scores” in statistics (Good 1952), but we prefer the term “surprisals” from information theory (Stone 2022), as it provides a measure of how surprising an observation is, given the probability density function f . Specifically, it takes large values for observations that are unlikely under f .

Let $S = -\log f(\mathbf{Y})$ be the random variable representing the surprisal of a random vector $\mathbf{Y}$ with density f , and let $G(s) = \mathbb{P}(S \leq s)$ be its distribution function. Then the surprisal probability of observation $\mathbf{y}_i$ is

$$p_i = \mathbb{P}(S \geq s_i) = 1 - G(s_i^-),$$

where $s_i = -\log f(\mathbf{y}_i)$, and $G(s_i^-) = \lim_{t \uparrow s_i} G(t)$ is the left limit of G at s_i . An observation is an anomaly if and only if $p_i < \alpha$.

1.2 Extreme value theory

The lookout algorithm determines which surprisals are unusually large by using a peaks-over-threshold approach (Pickands 1975, Coles 2001), and fitting a GPD to the upper tail of the observed surprisals. That is,

$$\mathbb{P}(S \leq s \mid S > u) = \frac{G(s) - G(u)}{1 - G(u)} \approx 1 - (1 + \xi(s - u)/\sigma_u)^{-1/\xi}, \quad s > u, \quad 1 + \xi(s - u)/\sigma_u > 0, \quad (2)$$

where u is a threshold chosen to be large enough that the conditional distribution of the excess $S - u$ over the threshold is well approximated by a GPD, $\sigma_u > 0$ is a scale parameter depending on the threshold, and $\xi \in \mathbb{R}$ is a shape parameter. The case $\xi = 0$ is read as the limiting case $1 - \exp(-(s - u)/\sigma_u)$.

1.3 Kernel density estimation

Given a set of observations $\mathbf{y}_i \in \mathbb{R}^m$ for $i \in \{1, \dots, n\}$, a kernel density estimate is defined as (Chacón & Duong 2018)

$$\hat{f}(\mathbf{y}) = \frac{1}{n} \sum_{i=1}^n |\mathbf{H}|^{-1/2} K(\mathbf{H}^{-1/2}(\mathbf{y} - \mathbf{y}_i)), \quad (3)$$

where K is a kernel function and $\mathbf{H}$ is a symmetric $m \times m$ positive-definite bandwidth matrix. When the observations are n independent draws from f , the accuracy of $\hat{f}$ is conventionally measured by the mean integrated squared error (MISE),

$$\text{MISE}(\hat{f}) = \mathbb{E} \left[\int (\hat{f}(\mathbf{u}) - f(\mathbf{u}))^2 d\mathbf{u} \right], \quad (4)$$

where the expectation is taken over that sample, and $\hat{f}$ is a *consistent estimator* of f if $\text{MISE}(\hat{f}) \rightarrow 0$ as $n \rightarrow \infty$. We consider matrices of the form $\mathbf{H} = ah_n^2 \mathbf{I}_m$ for a fixed constant $a > 0$, where h_n is a length scale computed from the sample, so that $|\mathbf{H}|^{1/2} = a^{m/2} h_n^m$.

For such a bandwidth matrix, $\hat{f}$ is a consistent estimator of f whenever

$$h_n \rightarrow 0 \quad (5)$$

and

$$nh_n^m \rightarrow \infty \quad (6)$$

in probability as $n \rightarrow \infty$, provided that f is square integrable and twice differentiable with bounded, continuous and square integrable second derivatives, and K is a square integrable, spherically symmetric density with a finite second moment (Chacón & Duong 2018, pp. 31–37). Neither version of the lookout algorithm has a bandwidth satisfying both conditions in general: under

heavy tails the original bandwidth need not satisfy (5), and the modified bandwidth never satisfies (6). In Section 2.3 we argue that a consistent estimate of f is not required for anomaly detection, and that the second behaviour is the appropriate one.

1.4 The lookout algorithm for anomaly detection

The *lookout* algorithm of Kandanaarachchi & Hyndman (2022), as implemented in the `lookout` package, has four inputs: the dataset $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, where each $\mathbf{y}_i \in \mathbb{R}^m$; a parameter $\beta \in (0, 1)$ such that the GPD is fitted to the largest $1 - \beta$ of the surprisals, with $\beta = 0.9$ following Bommier (2014); a parameter $\alpha \in (0, 1)$ signifying the cut-off threshold for determining an anomaly; and a logical variable *unitize* indicating whether to scale the data. We briefly outline the steps below.

Two details of Algorithm 1 differ from the published description. From version 0.1.4 of the `lookout` package, the search in step 4 is restricted to the death diameters at or above their median, so that a large gap among the shortest diameters is not chosen. Equation (5) of Kandanaarachchi & Hyndman (2022) gives the bandwidth matrix as $d_*^{2/m} \mathbf{I}_m$, whereas the implementation uses $d_*^2 \mathbf{I}_m$ in every dimension, as in step 5.

The lookout algorithm provides density-based anomaly detection, where anomalies are identified as points with low density in the space of the data. If there are no true anomalies in the data, the algorithm should identify a proportion α of the data as false positive “anomalies”.

Steps 2–4 of Algorithm 1 can be expressed in terms of the Euclidean minimum spanning tree T of the data. Components of the Vietoris-Rips complex merge when the shortest edge between them appears, which is Kruskal’s algorithm (Kruskal 1956), so the merges occur along the edges of T . The finite death diameters $d_1 \leq \dots \leq d_{n-1}$ are therefore the ordered edge lengths $e_{(1)} \leq \dots \leq e_{(n-1)}$ of T (Adams et al. 2020, Remark 1), and also the merge heights of the single linkage dendrogram (Gower & Ross 1969). In the rest of the paper we use the tree rather than the filtration, so that we can apply the limit theory for minimum spanning trees. The tree also gives a faster computation:

Algorithm 1 Lookout algorithm

Require: Data $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, proportion $\beta \in (0, 1)$, threshold $\alpha \in (0, 1)$, *unitize* $\in \{\text{True}, \text{False}\}$.

- 1: If *unitize*, scale the data to lie in the unit cube in $\mathbb{R}^m$.
 - 2: Compute the persistent homology barcode of the (possibly scaled) data cloud for dimension zero using the Vietoris-Rips diameter (Ghrist 2008).
 - 3: From the barcode, obtain the ordered finite death diameters $\{d_i\}_{i=1}^{n-1}$ for connected components, and their successive differences $\Delta d_i = d_{i+1} - d_i$.
 - 4: Let $d_* = d_{i_*}$ where $i_* = \arg \max\{\Delta d_i\}_{i=1}^{n-2}$; that is, d_* is the diameter corresponding to the lower point of the maximum Δd_i .
 - 5: Compute the kernel density estimates at the observations: $f_i = \hat{f}(\mathbf{y}_i)$, $i = 1, \dots, n$, where $\hat{f}$ is given by (3) with the kernel $K(\mathbf{u}) = \frac{m+2}{2 \cdot 5^{m/2} b_m} (1 - \|\mathbf{u}\|^2/5)_+$, b_m being the volume of the unit ball in $\mathbb{R}^m$, and $\mathbf{H} = d_*^2 \mathbf{I}_m$, so that the kernel has support radius $\sqrt{5} d_*$.
 - 6: Compute the leave-one-out estimates $f_{-i} = \frac{1}{n-1} (n f_i - |\mathbf{H}|^{-1/2} K(\mathbf{0}))$ for $i = 1, \dots, n$.
 - 7: Fit a GPD to the largest $1 - \beta$ of the surprisals $\{-\log f_i\}_{i=1}^n$, with location μ equal to the β sample quantile of the surprisals, obtaining estimates of the scale and shape parameters (σ, ξ) .
 - 8: Calculate the probability of each observation by applying the estimated GPD to the leave-one-out estimates: $p_i = (1 - \beta)(1 - \hat{F}(-\log f_{-i}))$, where $\hat{F}$ is the distribution function of the fitted GPD.
 - 9: If $p_i < \alpha$, declare $\mathbf{y}_i$ anomalous.
-

the original `lookout` package computed the death diameters from all $n(n-1)/2$ pairwise distances, while the algorithm described in [Section 2](#) computes the same values in $O(n \log n)$ time.

While [Kandanaarachchi & Hyndman \(2022\)](#) demonstrated excellent results in applying their algorithm to a variety of simulated and real datasets, we see several potential areas for improvement that we aim to address in this paper. First, elements of the algorithm are not robust to outliers, including the min-max scaling, and the use of the largest gap in the Rips death diameters (equivalently, edge lengths). Second, the diagonal bandwidth matrix $\mathbf{H}$ does not account for correlations between variables, which can lead to suboptimal performance if the data have strong correlations. Third, the behaviour of d_* as n grows is not known: it is not obvious that it shrinks to zero as in [\(5\)](#), particularly for distributions with heavy tails. Finally, the GPD estimation made minimal assumptions about the distribution of the surprisal values, but we know that these are bounded above and below, and that fact can be used to improve the estimated GPD.

1.5 The modified lookout algorithm

As a result of these considerations, we propose three modifications to the lookout algorithm, described in detail in [Section 3](#).

1. Rather than use a min-max scaling of the data, we standardize the data using a robust covariance estimator, so that the transformed data have approximately uncorrelated coordinates with unit variance.
2. Rather than setting the bandwidth using the edge length at the lower end of the largest gap between consecutive minimum spanning tree edge lengths, we set it using an upper quantile of those edge lengths. This avoids the problem that a few anomalies at the top of the edge length distribution can determine the largest gap and hence the bandwidth. It also gives the bandwidth the order $n^{-1/m}$ of the spacing between neighbouring observations, for any density, at which nh_n^m stays bounded and [\(6\)](#) fails.

3. The estimated GPD shape parameter is constrained to be non-positive, which improves the stability of the GPD fitting.

1.6 Other uses of the minimum spanning tree

Minimum spanning trees have been used for anomaly detection for many years. [Rohlf \(1975\)](#) gave the classical minimum spanning tree test for multivariate outliers. When $m = 1$, the edges of the minimum spanning tree are the gaps between consecutive order statistics, so the edge lengths of the tree generalize the gaps of the univariate gap test; Rohlf tests the largest squared edge length, divided by the mean squared edge length, with critical values derived from a gamma distribution for the squared edge lengths. However, [Caroni & Prescott \(1995\)](#) argue that this is not a useful formal test.

Related uses of the tree include single linkage detectors, which exploit the equivalence between the minimum spanning tree and single linkage clustering ([Gower & Ross 1969](#)) to cut the dendrogram and declare the resulting small clusters anomalous ([Jiang et al. 2001](#)). [Juszczak et al. \(2009\)](#) build a one-class classifier from the tree of a training sample, and [Ahmed et al. \(2019\)](#) replace Euclidean distances by distances along the tree in a local outlier score. These detectors identify anomalies from the tree. We use it only to choose the bandwidth, and anomalies are identified by the leave-one-out kernel density estimate and the fitted generalized Pareto distribution, as in [Algorithm 1](#), so a long edge of T does not by itself make its endpoint an anomaly, and an observation can be anomalous without being incident to a long edge.

The tree has also been used to choose the bandwidth of a kernel density estimate, with the aim of estimating the density over its whole support. [Chaudhuri et al. \(1996\)](#) show that the resulting estimate with a uniform kernel is asymptotically unbiased on bounded sets, and [Sreevani & Murthy \(2016\)](#) take the bandwidth to be a function of the total length of the tree, with outlier rejection among the applications. The total length is determined by the bulk of the data. In contrast, we

choose our bandwidth for estimating the density at isolated observations (Section 2.3), and our upper quantile of the edge lengths is determined by the sparsest part of the sample.

1.7 Outline of the paper

In Section 2 we define the bandwidth as a quantile of the edge lengths of the minimum spanning tree, establish its order for any density, and show that the longest edges, which the original algorithm typically uses in its bandwidth, can grow without bound. Section 3 gives the modified algorithm. Section 4 compares it with the original and with four other methods in five simulation experiments, and examines the sensitivity of the results to the bandwidth. Section 5 applies both versions to two real datasets, and Section 6 discusses the results.

2 Bandwidth selection from the minimum spanning tree

2.1 The bandwidth rule

Let T denote the Euclidean minimum spanning tree of $\{\mathbf{y}_1, \dots, \mathbf{y}_n\} \subset \mathbb{R}^m$: the connected graph on the n observations whose total edge length is smallest. It has $n - 1$ edges, and it is unique provided the pairwise distances are distinct, which holds almost surely when the observations are drawn from a continuous density. When placed in increasing order, the edge lengths are denoted by $e_{(1)} \leq \dots \leq e_{(n-1)}$.

Definition 3 (Minimum spanning tree bandwidth). For $\gamma \in (0, 1)$, the *minimum spanning tree bandwidth* is the γ quantile of the edge lengths of T ,

$$h_n = e_{(\lceil \gamma(n-1) \rceil)}. \quad (7)$$

We take $\gamma = 0.98$ throughout unless stated otherwise. In computation we use the approximately median-unbiased quantile estimator of Hyndman & Fan (1996), their type 8, which has better finite sample properties than the order statistic. Its value lies between $e_{(k-1)}$ and $e_{(k+1)}$, where

$k = \lceil \gamma(n-1) \rceil$, and [Proposition 1](#) below holds for it by the same argument. The kernel density estimate (3) is then computed with the bandwidth matrix

$$\mathbf{H} = ah_n^2 \mathbf{I}_m, \quad (8)$$

where $a = m + 4$ and K is the Epanechnikov kernel $K(\mathbf{u}) = \frac{m+2}{2b_m}(1 - \|\mathbf{u}\|^2)_+$, with b_m the volume of the unit ball in $\mathbb{R}^m$. The kernel then has support radius $\sqrt{m+4}h_n$, and h_n is its per-coordinate standard deviation.

Equation (7) has an interpretation in terms of single linkage clustering. Cutting the single linkage dendrogram at height h_n leaves about $(1 - \gamma)n$ clusters. With $\gamma = 0.98$, h_n is the height at which 2% of the merges remain to be made, so at most about 2% of the observations are still isolated.

To compute h_n , we use the dual-tree Borůvka algorithm of [March et al. \(2010\)](#), as implemented in `mlpack` ([Singh Parihar et al. 2026](#)), which builds the exact Euclidean minimum spanning tree in $O(n \log n)$ time under the same assumptions that make dual-tree nearest-neighbour search efficient.

[Fig. 1](#) shows the rule on a sample of 300 standard normal observations in $\mathbb{R}^2$ with three anomalies added. The left panel shows the minimum spanning tree, with the edges longer than h_n drawn in colour; the middle panel shows the single linkage dendrogram with the cut at h_n ; the right panel shows the kernel density estimate computed with the bandwidth matrix (8).

2.2 The order of the quantile edge length

We study the quantile through its relationship to single linkage clustering. For $r > 0$, let $C_n(r)$ be the number of connected components of the random geometric graph that joins every pair of observations at distance at most r . Kruskal’s algorithm ([Kruskal 1956](#)) builds T by adding edges in increasing order of length, so the edges of T of length at most r form a spanning forest of that graph, with one tree for each component ([Gower & Ross 1969](#)). There are therefore $n - C_n(r)$ such edges. Since h_n is the $\lceil \gamma(n-1) \rceil$ th smallest edge length, $h_n \leq r$ exactly when at least $\lceil \gamma(n-1) \rceil$ edges have length at most r , that is, when $C_n(r) \leq n - \lceil \gamma(n-1) \rceil$.

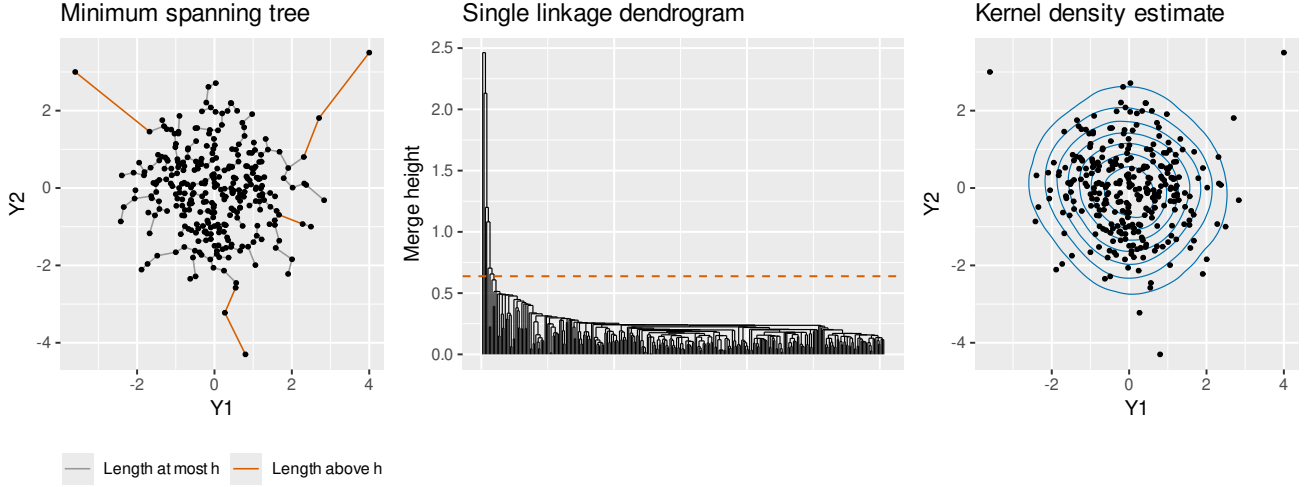


Figure 1: *The minimum spanning tree bandwidth for 300 standard normal observations in $\mathbb{R}^2$ and three anomalies. Left: the minimum spanning tree; edges longer than h_n are shown in colour. Middle: the single linkage dendrogram, with the dashed line at height h_n . Right: contours of the kernel density estimate with bandwidth matrix (8).*

Component counts of random geometric graphs obey a law of large numbers (Penrose 2003, Theorem 13.25), which Mitsche & Penrose (2021) extend to points drawn from any density. The limit involves a Poisson process. For $\lambda > 0$, let $\mathcal{P}_\lambda$ be a homogeneous Poisson process of intensity λ on $\mathbb{R}^m$, join each pair of points of $\mathcal{P}_\lambda \cup \{\mathbf{0}\}$ at distance at most 1, let $\mathcal{C}_\lambda$ be the component containing the origin and $|\mathcal{C}_\lambda|$ its number of points, and put $\kappa(\lambda) = \mathbb{E}[1/|\mathcal{C}_\lambda|]$, with $1/|\mathcal{C}_\lambda| = 0$ when the component is infinite. A component with s points contributes $1/s$ at each of its points, so the sum over all points of the reciprocal of the size of their component is the number of components. Thus $\kappa(\lambda)$ is the expected number of components per point in a large sample when points of intensity λ are joined at unit distance.

Proposition 1. *Let $\mathbf{y}_1, \dots, \mathbf{y}_n$ be sampled independently from a density f on $\mathbb{R}^m$, and fix $\gamma \in (0, 1)$. Then $n^{1/m}h_n \rightarrow q_\gamma(f)$ in probability as $n \rightarrow \infty$, where $q_\gamma(f) \in (0, \infty)$ is the unique solution q of*

$$\int_{\mathbb{R}^m} \kappa(q^m f(\mathbf{x})) f(\mathbf{x}) d\mathbf{x} = 1 - \gamma. \quad (9)$$

Moreover $t_- \leq q_\gamma(f) \leq t_+$, where t_- and t_+ are the unique positive solutions t of

$$\int_{\mathbb{R}^m} \exp(-b_m t^m f(\mathbf{x})) f(\mathbf{x}) d\mathbf{x} = 1 - \gamma \quad \text{and} \quad \int_{\mathbb{R}^m} (1 - \exp(-b_m t^m f(\mathbf{x}))) d\mathbf{x} = (1 - \gamma) b_m t^m$$

respectively, and b_m is the volume of the unit ball in $\mathbb{R}^m$.

The proof is in the online Appendix. No smoothness, boundedness or tail condition on f is required. Equation (9) is the single linkage interpretation in the limit: after rescaling distances by $n^{1/m}$, $q_\gamma(f)$ is the scale at which a large sample from f has $1 - \gamma$ components per observation.

For the bandwidth matrix (8), [Proposition 1](#) has the following consequence.

Corollary 1. *Under the conditions of [Proposition 1](#), let $\mathbf{H} = (m + 4)h_n^2 \mathbf{I}_m$ with $h_n = e_{(\lceil \gamma(n-1) \rceil)}$. Then $n|\mathbf{H}|^{1/2} \rightarrow (m + 4)^{m/2} q_\gamma(f)^m$ in probability as $n \rightarrow \infty$, and the limit is finite and positive.*

Proof. $|\mathbf{H}|^{1/2} = (m + 4)^{m/2} h_n^m$, so $n|\mathbf{H}|^{1/2} = (m + 4)^{m/2} (n^{1/m} h_n)^m$, and [Proposition 1](#) gives $n^{1/m} h_n \rightarrow q_\gamma(f) \in (0, \infty)$ in probability. $\square$

The quantile bandwidth therefore satisfies (5) but not (6), since nh_n^m converges to a positive constant. The integrated variance of $\hat{f}$ is approximately $n^{-1}|\mathbf{H}|^{-1/2}R(K)$, where $R(K) = \int K(\mathbf{u})^2 d\mathbf{u}$, so it does not vanish, and the estimator is not MISE-consistent. The expected number of observations inside the kernel support at a point $\mathbf{x}$ is asymptotically $nb_m|\mathbf{H}|^{1/2}f(\mathbf{x})$, which by [Corollary 1](#) converges to $b_m(m + 4)^{m/2}q_\gamma(f)^m f(\mathbf{x})$. The number of observations in the window is therefore bounded as $n \rightarrow \infty$, and is proportional to the local density. In [Section 2.3](#) we argue that a bandwidth of order $n^{-1/m}$ is appropriate for anomaly detection.

2.3 The bandwidth and the MISE

An asymptotic approximation to the MISE (4) is given by [Chacón & Duong \(2018\)](#):

$$\text{AMISE}(\hat{f}) = n^{-1}|\mathbf{H}|^{-1/2}R(K) + \frac{1}{4}\mu_2(K)^2 \int \{\text{tr}(\mathbf{H}D^2 f(\mathbf{u}))\}^2 d\mathbf{u}, \quad (10)$$

where the constant $\mu_2(K)$ is defined by $\int \mathbf{u}\mathbf{u}^T K(\mathbf{u}) d\mathbf{u} = \mu_2(K)\mathbf{I}_m$, and D^2f is the Hessian of f . Writing $\mathbf{H} = (m+4)h^2\mathbf{I}_m$ as in (8), so that h is in the same units as h_n , this is

$$\text{AMISE} = n^{-1}(m+4)^{-m/2}h^{-m}R(K) + \frac{(m+4)^2}{4}\mu_2(K)^2h^4 \int \{\Delta f(\mathbf{u})\}^2 d\mathbf{u}, \quad (11)$$

minimized at $h_{\text{opt}} \asymp n^{-1/(m+4)}$, with MISE of order $n^{-4/(m+4)}$. By Proposition 1, $h_n \asymp n^{-1/m}$. The ratio is

$$\frac{h_n}{h_{\text{opt}}} \asymp n^{-4/\{m(m+4)\}},$$

which tends to zero for every $m \geq 1$. Asymptotically, the minimum spanning tree bandwidth undersmooths relative to MISE in every dimension: at $h \asymp n^{-1/m}$ the bias term of (11) is $O(n^{-4/m})$ while the variance term converges to a positive constant. However, $h_{\text{opt}} = cn^{-1/(m+4)}$, where c does not depend on n , so h_n/h_{opt} is approximately $(q_\gamma(f)/c)n^{-4/\{m(m+4)\}}$. For $m \geq 3$ the exponent $4/\{m(m+4)\}$ is small. For the standard normal density, c is between 0.9 and 1 and $q_\gamma(f)/c$ is between about 6 and 9 when $m = 3$ or $m = 5$, so the asymptotic comparison is misleading at practical sample sizes. Table 1 reports $n^{1/m}h_n$ for standard normal samples with $\gamma = 0.98$, the ratio of h_n to the bandwidth h_{opt} that minimizes (11) for the Epanechnikov kernel and this density, and the expected number of observations inside the kernel support at the mode. The scaled quantile changes little with n . The ratio h_n/h_{opt} is below one only for $m = 2$ and $n \geq 4000$. For $m = 3$ it is still 1.42 at $n = 16000$, and for $m = 5$ it is about three at all three sample sizes and decreases as $n^{-4/45}$. In moderate dimensions the rule therefore smooths the bulk of the data more than the MISE-optimal bandwidth does, and for $m = 5$ the window at the mode contains most of the sample.

The MISE integrates the squared error over $\mathbb{R}^m$. The pointwise variance of $\hat{f}$ is approximately proportional to f and the pointwise bias to Δf , so both are largest in the bulk of the distribution, and h_{opt} depends on f through $\int \{\Delta f(\mathbf{u})\}^2 d\mathbf{u}$, which is dominated by the bulk. Anomalies are identified from the values of $\hat{f}$ at observations where f is small, and from the upper tail of the surprisals, $-\log \hat{f}(\mathbf{y}_i)$, so the relevant spacing is that of the observations in the tails. By the spacing in the tails we mean the distance between neighbouring observations in the low-density region of f ,

Table 1: *The minimum spanning tree bandwidth for standard normal samples in $\mathbb{R}^m$ with $\gamma = 0.98$, averaged over ten samples. h_{opt} minimizes the AMISE (11) for the Epanechnikov kernel and this density. The last column is the expected number of observations inside the kernel support at the mode, $n \mathbb{P}(\|\mathbf{Y}\| \leq \sqrt{m+4} h_n)$.*

m	n	$n^{1/m} h_n$	h_n/h_{opt}	Observations in window at mode
2	1000	10.6	1.08	290
2	4000	12.1	0.77	410
2	16000	12.3	0.50	450
3	1000	7.4	2.12	720
3	4000	8.2	1.79	1,600
3	16000	8.5	1.42	2,400
5	1000	5.8	3.51	1,000
5	4000	6.1	3.24	3,900
5	16000	6.3	2.98	13,000

excluding the most isolated few observations, whose spacing can grow without bound (Section 2.4). A bandwidth determined by the bulk may be either too small or too large for the tails. If it is smaller than the spacing in the tails, as h_{opt} is for $m \geq 3$ in Table 1, then any observation with no other observation inside its kernel support has a leave-one-out estimate of exactly zero, and the isolated observations cannot be ranked. If it is larger, as h_{opt} is for $m = 2$ and large n , then the estimate at an isolated observation includes contributions from distant observations, and the observation no longer appears isolated. The quantile rule sets the bandwidth from the spacing in the tails. By the single linkage interpretation of (7), h_n is the scale at which all but about 2% of the observations are in clusters of two or more, so the kernel support around any observation other than the most isolated few contains at least one other observation. For $m \geq 3$ the estimate

in the bulk is oversmoothed at practical sample sizes and undersmoothed in the limit. For $m = 2$ it is already undersmoothed at $n = 4000$. This has little effect on anomaly detection, because the estimates in the bulk are large in either case, and those observations are not identified as anomalies.

We therefore do not require consistency or an optimal rate. We require that h_n has the same order as the spacing in the tails. At a point $\mathbf{x}$ with $f(\mathbf{x}) > 0$, the distance to the nearest observation is of order $\{nf(\mathbf{x})\}^{-1/m}$, so at any fixed density level the spacing is of order $n^{-1/m}$. [Proposition 1](#) shows that $n^{1/m}h_n$ converges to $q_\gamma(f)$, the rescaled distance at which a large sample from f has $1 - \gamma$ components per observation, so that at most about 2% of the observations are still isolated. In [Section 4.6](#) we examine the sensitivity of the results to the bandwidth, by scaling h_n up and down and by replacing it with h_{opt} .

2.4 A fixed quantile versus the largest gap

The original rule of [Algorithm 1](#) sets the bandwidth using the edge length at the lower end of the largest gap between consecutive edge lengths. Under heavy tails, that gap typically lies among the few longest edges, where [Proposition 1](#) says nothing and the edge lengths can grow without bound.

Counterexample 1. *Let the observations be sampled independently from the density $f(\mathbf{x}) = c_m/(1 + \|\mathbf{x}\|^{m+1})$ on $\mathbb{R}^m$, $m \geq 1$, and put $\omega_n = n - \lfloor n^{1/4}/4 \rfloor$, which is at most $n - 1$ once $n \geq 256$. Then $\mathbb{P}(e_{(\omega_n)} > n^{1/8}) \rightarrow 1$ as $n \rightarrow \infty$. In particular $e_{(\omega_n)}$, and every longer edge of T , tends to infinity in probability.*

The proof is in the online Appendix. With probability tending to one, more than $n^{1/4}/4$ observations lie beyond radius $R_n \asymp n^{3/4}$, and no two observations beyond radius $R_n - n^{1/8}$ are within distance $n^{1/8}$ of each other. Each observation beyond R_n is then a component on its own when observations are joined at distance $n^{1/8}$, so $C_n(n^{1/8}) > n - \omega_n$: fewer than ω_n edges of T have length at most $n^{1/8}$, and $e_{(j)} > n^{1/8}$ for every $j \geq \omega_n$. [Adler et al. \(2014\)](#) call this behaviour

crackle. Under heavy tails, isolated observations occur at increasing distances from the origin, and each of them contributes one of the longest edges of T . These edges lie above any fixed quantile once $n^{1/4}/4 < (1 - \gamma)n$, and [Proposition 1](#) gives h_n of order $n^{-1/m}$ for this density as for any other. Under this density the edge lengths $e_{(j)}$ with $j \geq \omega_n$ tend to infinity, so any bandwidth proportional to one of them violates (5). The largest-gap rule uses $e_{(j)}$ in its bandwidth, where $e_{(j+1)} - e_{(j)}$ is the largest gap between consecutive edge lengths, so its bandwidth violates (5) whenever $j \geq \omega_n$, which is typical under heavy tails.

The limiting distribution of the longest edge $e_{(n-1)}$ is known for uniform points in the unit square ([Penrose 1997](#)), for Gaussian points ([Penrose 1998](#)), and for a range of distributions on $\mathbb{R}^2$ with unbounded support ([Hsing & Rootzén 2005](#)). In the Gaussian case in $\mathbb{R}^m$ with $m \geq 2$, the longest edge is of order $\log \log n / \sqrt{\log n}$, which tends to zero much more slowly than $n^{-1/m}$. The largest-gap bandwidth therefore has order $n^{-1/m}$ only if the largest gap lies well below the top of the edge-length distribution.

3 Lookout algorithm v2

In light of the preceding theoretical considerations, we propose three modifications to the lookout algorithm, described in this section.

3.1 Standardization

The *minimum covariance determinant* estimator provides a robust estimate of Σ , the covariance matrix of $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, by searching for the subsample of a given size whose sample covariance matrix has the smallest determinant, followed by a reweighting step ([Rousseeuw & Driessen 1999](#)). We take that subsample to be of size $0.9n$. This has a lower breakdown point than the usual choice of $n/2$, but anomalies are assumed to be rare, and the larger subsample gives a more

efficient estimate of Σ for the non-anomalous observations. Let

$$\hat{\Sigma}^{-1} = \mathbf{U}'\mathbf{U} \quad (12)$$

denote the Cholesky decomposition of its inverse. Then, the data can be scaled by $\mathbf{U}$, giving $\mathbf{z}_i = \mathbf{U}(\mathbf{y}_i - \mathbf{m})$, where $\mathbf{m}$ is the vector of medians of $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$. The covariance matrix of the scaled data $\{\mathbf{z}_1, \dots, \mathbf{z}_n\}$ is (approximately) the identity matrix. This can be thought of as both scaling and rotating the data; it is the multivariate equivalent of computing robust z-scores.

We prefer this to the min-max scaling used in the original lookout algorithm. The minimum covariance determinant estimate is robust to outliers, whereas the minimum and maximum of each variable are often determined by the anomalies themselves, and the transformation removes the correlations between variables. The kernel in [Algorithm 2](#) has a covariance matrix proportional to the identity, and the covariance matrix of the standardized data is approximately the identity, so the kernel has the same orientation as the data. This is not necessary, but it improves the efficiency of the kernel density estimate ([Chacón & Duong 2018](#)).

3.2 Bandwidth matrix selection

Let $e_{(1)} \leq \dots \leq e_{(n-1)}$ denote the ordered edge lengths of the Euclidean minimum spanning tree of the standardized data $\mathbf{z}_1, \dots, \mathbf{z}_n$, and let $h_n = e_{(\lceil \gamma(n-1) \rceil)}$ be the bandwidth of [Definition 3](#). The kernel density estimate is computed from the standardized data with the bandwidth matrix (8), so that by [Corollary 1](#) the expected number of observations inside the kernel window is bounded as $n \rightarrow \infty$, which [Section 2.3](#) argues is appropriate for anomaly detection. When the proportion of anomalies is small, they have little effect on the estimate at the non-anomalous observations. An equivalent estimate for the original data would use $\mathbf{H} = (m+4)h_n^2\hat{\Sigma}$; we work with the standardized data throughout.

3.3 Bounds for kernel density estimates

The surprisals to which the GPD is fitted are bounded, which constrains the shape parameter. Let K be a kernel with $0 \leq K(\mathbf{u}) \leq K(\mathbf{0}) = K_0$ for all $\mathbf{u}$, and let $\hat{f}$ be the estimate (3) with bandwidth matrix $\mathbf{H}$. The term $i = j$ of the sum contributes $|\mathbf{H}|^{-1/2} K_0/n$ and no term exceeds that, so

$$\frac{K_0 |\mathbf{H}|^{-1/2}}{n} \leq \hat{f}(\mathbf{y}_j) \leq K_0 |\mathbf{H}|^{-1/2}, \quad j = 1, \dots, n,$$

and the surprisals $-\log \hat{f}(\mathbf{y}_j)$ lie in the interval $[\frac{1}{2} \log |\mathbf{H}| - \log K_0, \frac{1}{2} \log |\mathbf{H}| + \log(n/K_0)]$, whose endpoints depend on the sample only through n and $\mathbf{H}$. For the Epanechnikov kernel $K(\mathbf{u}) = \frac{m+2}{2b_m} (1 - \|\mathbf{u}\|^2)_+$, where $u_+ = \max(0, u)$, $K_0 = (m+2)/(2b_m)$.

A GPD has a bounded upper tail when its shape parameter is negative, an exponential tail when it is zero, and a polynomial tail when it is positive. A polynomial tail is inconsistent with bounded surprisals, so when we fit the GPD to the largest $1 - \beta$ of $\{-\log \hat{f}(\mathbf{y}_j)\}_{j=1}^n$ we constrain its shape parameter to be non-positive, which improves the stability of the fit.

The GPD is fitted under two approximations. The surprisals are computed from $\hat{f}$ rather than from f , and they are evaluated at the observations rather than at independent draws, so the values to which the GPD is fitted are neither mutually independent nor distributed exactly as S in [Section 1](#).

3.4 Modified lookout algorithm

[Algorithm 2](#) summarises the modified algorithm.

As with the original lookout algorithm, the values of p_i can also act as anomaly scores, with lower values indicating more anomalous points. The probability p_i in step 7 estimates the probability of a surprisal at least as large as $-\log f_{-i}$. When $-\log f_{-i}$ exceeds μ , this is the probability $1 - \beta$ of exceeding μ multiplied by the conditional exceedance probability from the fitted GPD; otherwise $p_i = 1 - \beta$. In step 6 the implementation fits the GPD without constraint and, if the estimated shape parameter is positive, refits it with the shape parameter fixed at zero.

Algorithm 2 Modified lookout algorithm

Require: Data $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, proportion $\beta \in (0, 1)$, threshold $\alpha \in (0, 1)$, and quantile level $\gamma \in (0, 1)$.

- 1: Optional scaling: calculate the minimum covariance determinant estimate $\hat{\Sigma}$ and the Cholesky decomposition (12), and centre, rotate and scale the data using $\mathbf{U}$, giving $\mathbf{z}_i = \mathbf{U}(\mathbf{y}_i - \mathbf{m})$ for $i = 1, \dots, n$, where $\mathbf{m}$ is the vector of medians of $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$.
 - 2: Compute the Euclidean minimum spanning tree of $\{\mathbf{z}_1, \dots, \mathbf{z}_n\}$ (set $\mathbf{z}_i = \mathbf{y}_i$ if no scaling is applied), and obtain its ordered edge lengths $e_{(1)} \leq \dots \leq e_{(n-1)}$.
 - 3: Let h_n be the γ sample quantile of $\{e_{(i)}\}_{i=1}^{n-1}$, computed with the type 8 estimator of Hyndman & Fan (1996).
 - 4: Compute the kernel density estimates at the scaled observations: $f_i = \hat{f}(\mathbf{z}_i)$, $i = 1, \dots, n$, where $\hat{f}$ is given by (3) applied to $\mathbf{z}_1, \dots, \mathbf{z}_n$ with the Epanechnikov kernel $K(\mathbf{u}) = \frac{m+2}{2b_m}(1 - \|\mathbf{u}\|^2)_+$ and $\mathbf{H} = (m+4)h_n^2\mathbf{I}_m$.
 - 5: Compute the leave-one-out estimates $f_{-i} = \frac{1}{n-1}(nf_i - |\mathbf{H}|^{-1/2}K(\mathbf{0}))$ for $i = 1, \dots, n$.
 - 6: Fit a GPD to the largest $1 - \beta$ of the surprisals $\{-\log f_i\}_{i=1}^n$, with location μ equal to the β sample quantile of the surprisals, constraining the shape parameter to be non-positive, and obtaining estimates of the scale and shape parameters (σ, ξ) .
 - 7: Calculate the probability of each observation by applying the estimated GPD to the leave-one-out estimates: $p_i = (1 - \beta)(1 - \hat{F}(-\log f_{-i}))$, where $\hat{F}$ is the distribution function of the fitted GPD.
 - 8: If $p_i < \alpha$, declare $\mathbf{y}_i$ anomalous.
-

The default values are $\alpha = 0.01$, $\beta = 0.90$ and $\gamma = 0.98$, with scaling applied. They are used in all experiments reported here unless stated otherwise.

Step 4 evaluates (3) at each of the n observations. The implementation finds the observations inside the kernel support of each observation by a fixed-radius search, so the computational cost is proportional to the number of such pairs. Table 1 shows that in three or more dimensions the support around an observation in the bulk of the data can hold a large fraction of the sample, in which case the cost is of order n^2 at these sample sizes, and the exact sum is practical only for n up to about 10^4 (Table 2). For larger n the implementation instead sums the kernel over the k observations nearest to each observation, including the observation itself, where k is between 100 and 500 depending on n . This underestimates $\hat{f}$ in the bulk. The observations relevant to anomaly detection have fewer than k observations inside their kernel support, so their estimates are unchanged. The truncated sum is used in Experiment 2, in Experiment S2 of the online Appendix, and for the wine reviews data in Section 5.

Table 2 gives the time taken by the tree and by the whole of Algorithm 2 for standard normal samples, on a desktop computer with an Intel Core Ultra 9 285 processor. The tree takes under four seconds in every case except $m = 10$ at $n = 10^5$, where it takes a minute. In Experiments 3 to 5 of Section 4, where $n \leq 805$, a run of lookout takes about 0.1 seconds or less; KDEOS takes about the same, stray and HDoutliers less, and RDOS one to five seconds.

4 Numerical experiments

We now evaluate the new version of lookout in five simulation experiments. Experiments 1 and 2 compare it against the original version, Experiments 3 to 5 compare both versions with four other anomaly detection methods, and Section 4.6 examines the sensitivity of the results to the bandwidth. Two further comparisons of the two versions, a set of showcase examples, and the details of Section 4.6 are in the online Appendix. Both versions are run in the current implementation, version 2.1.0 of the lookout package (Kandanaarachchi et al. 2026).

Table 2: Seconds taken by the minimum spanning tree and by the whole of [Algorithm 2](#) for standard normal samples in $\mathbb{R}^m$, on a desktop computer with an Intel Core Ultra 9 285 processor. The exact leave-one-out estimate uses a fixed-radius neighbour search and was run for $n \leq 10^4$, and up to $n = 10^5$ in two dimensions; the truncated estimate sums each kernel over at most 500 nearest neighbours.

m	n	Tree	Total, exact KDE	Total, truncated KDE
2	10^3	0.012	0.045	0.062
5	10^3	0.009	0.074	0.076
10	10^3	0.043	0.14	0.14
2	10^4	0.023	0.28	0.16
5	10^4	0.16	2.9	0.47
10	10^4	2	5.1	3.2
2	10^5	0.29	1.4	5.5
5	10^5	2.3		18
10	10^5	60		147
2	10^6	3.3		99

The four other methods, also used for comparison by [Kandanaarachchi & Hyndman \(2022\)](#), are HDoutliers ([Wilkinson 2017](#)) and *stray* ([Talagala et al. 2021](#)), which apply extreme value theory to nearest-neighbour distances, and KDEOS ([Schubert et al. 2014](#)) and RDOS ([Tang & He 2017](#)), which are based on kernel density estimates. HDoutliers and *stray* are run with $\alpha = 0.01$ (the default for *stray*), and KDEOS and RDOS with their default settings. Experiments 1 and 2 report the true and false positive rates of the two versions of lookout. HDoutliers and *stray* return labels, so in Experiments 3 to 5 we compare them with lookout by the Fmeasure and Gmean,

$$\text{Fmeasure} = \frac{2 \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \quad \text{Gmean} = \sqrt{\text{Sensitivity} \times \text{Specificity}},$$

where Precision = $TP/(TP + FP)$, Recall = Sensitivity = $TP/(TP + FN)$ and Specificity = $TN/(TN + FP)$, with TP, FP, TN and FN the numbers of true positives, false positives, true negatives and false negatives. KDEOS and RDOS return anomaly scores, so we compare them with the scores p_i of lookout by the area under the ROC curve (AUC), the curve of sensitivity against one minus specificity over all thresholds.

4.1 Experiment 1: Different anomaly rates for Gamma distribution

The non-anomalous observations have independent gamma coordinates in $\mathbb{R}^2$ with shape 2 and rate 2, and the anomalies independent gamma coordinates with shape 2 and rate $r \in \{0.1, 0.2, \dots, 1\}$. Each replication has 500 non-anomalies and 10 anomalies, and there are 10 replications for each value of r .

Fig. 2 shows the data and results. As r increases the anomalies become more similar to the non-anomalies, so the true positive rate falls. The new version has a higher true positive rate than the original at every value of r below 1, with the largest difference at $r = 0.1$ (0.97 against 0.19), where the anomalies are farthest from the non-anomalous observations. Both versions have false positive rates of at most 0.012.

4.2 Experiment 2: Anomalies on the boundary as n increases

The non-anomalous observations have independent $\mathcal{N}(0, 1)$ coordinates in $\mathbb{R}^2$, with $n \in \{1000, 2000, \dots, 10000\}$, and there are $0.005n$ anomalies in a ring just beyond most of them. The anomaly means μ_i lie on the circle of radius $2.2\sqrt{2}$ about the origin, with $\mu_{i1} \sim \mathcal{U}(-2.2\sqrt{2}, 2.2\sqrt{2})$ and μ_{i2} negative for a randomly chosen half of the anomalies, and $Y_{ij} \sim \mathcal{N}(\mu_{ij}, 0.1^2)$ for $j = 1, 2$. There are 10 replications for each n , and for both versions the kernel density estimates use the truncated sum described in Section 3, with 100 neighbours.

Fig. 3 shows the data of one replication and the results. The false positive rates are about 0.012 for the new version and 0.010 for the original. The two versions have similar true positive

rates at every n , between about 0.55 and 0.8. The online Appendix repeats this experiment with gamma-distributed observations (Experiment S2); there the new version has a true positive rate at least as high as the original at every n , and a false positive rate of about 0.012 against 0.008.

4.3 Experiment 3: Normal distributions moving out in each iteration

The sample has 405 points in $\mathbb{R}^6$, of which five are anomalies. The non-anomalies are $\mathcal{N}(0, 1)$ in every dimension. The anomalies differ from the non-anomalies only in the first dimension, where they are $\mathcal{N}(1.5 + 0.5i, 0.2^2)$ at iteration $i \in \{1, \dots, 10\}$, so they move further from the non-anomalous points at each iteration. Each iteration is repeated 10 times.

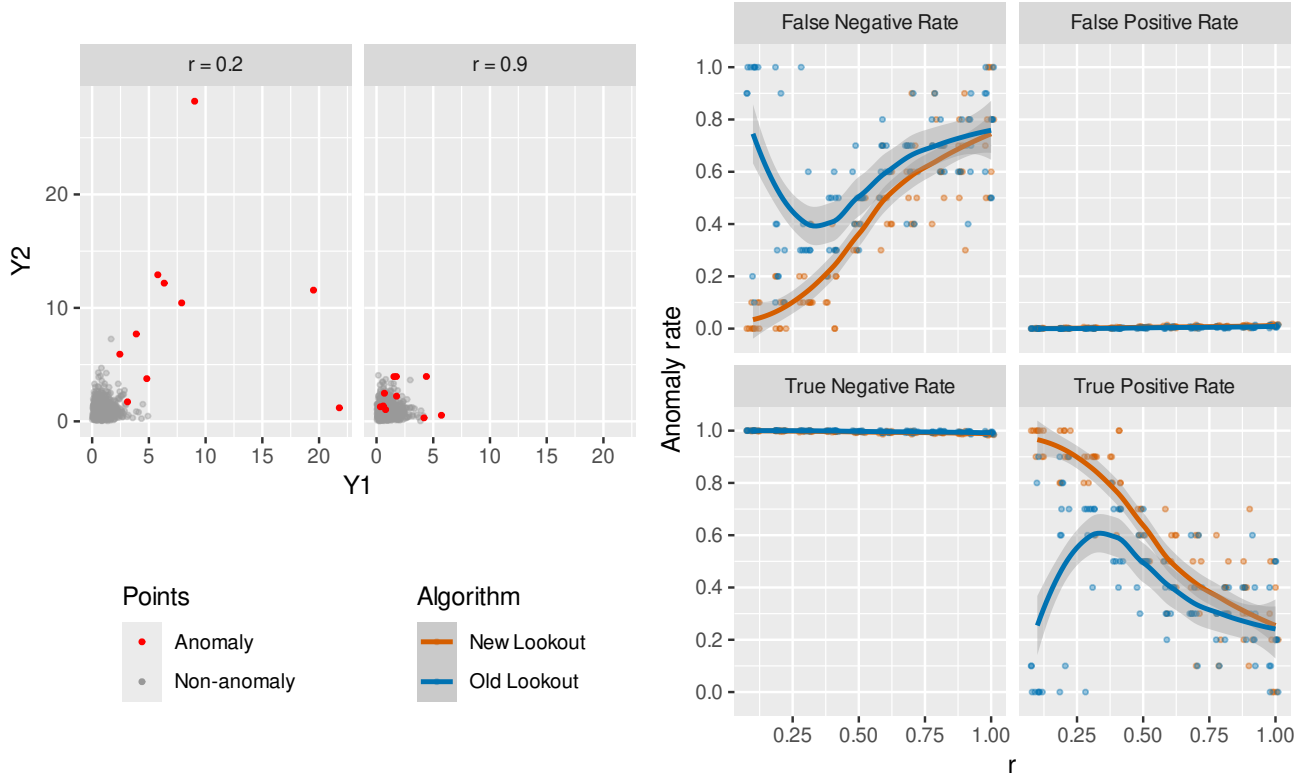


Figure 2: *Experiment 1. Left: data with $r = 0.2$ and $r = 0.9$; anomalies with lower values of r are easier to identify. Right: error rates of the two versions of lookout, with points jittered horizontally.*

Fig. 4 shows the data at the third and ninth iterations, and the results. By the ninth iteration the anomalies have separated from the other points in the Y_1 direction, but they remain mixed with them on the Y_3Y_4 plane. The new version has higher Fmeasure and Gmean than the original from the fourth iteration, and the two versions have similar AUC. RDOS has the highest AUC in the first three iterations; from the fourth, both versions of lookout are at least as good as the four other methods on every metric.

4.4 Experiment 4: Annulus in $\mathbb{R}^3$

The sample has 805 points in $\mathbb{R}^3$, of which five are anomalies. On the Y_1Y_2 plane the non-anomalies lie in an annulus, with uniform angle and $\mathcal{N}(7, 1)$ radius, and $Y_3 \sim \mathcal{U}(0, 1)$ for all points. At iteration $i \in \{1, \dots, 10\}$ the anomalies have $Y_1 \sim \mathcal{N}(5.5 - 0.5i, 0.1^2)$ and $Y_2 \sim \mathcal{N}(0, 0.1^2)$, so

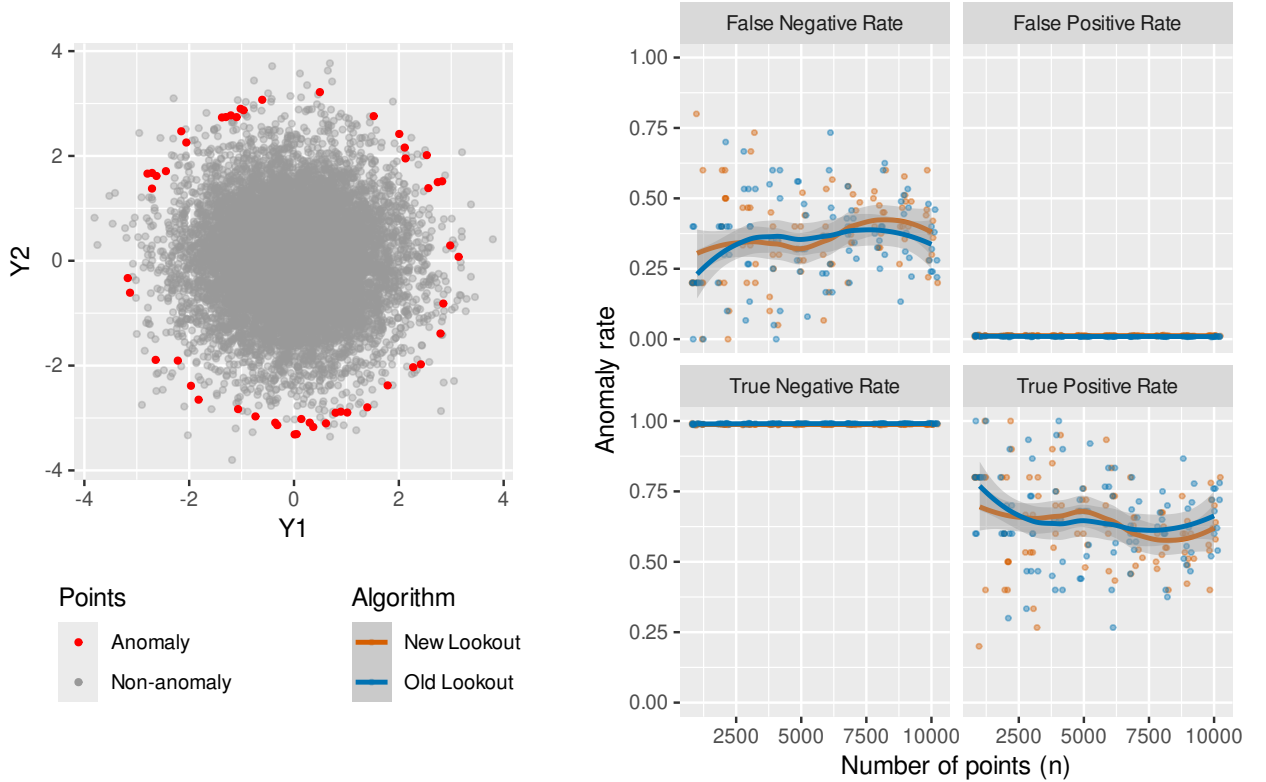


Figure 3: *Experiment 2. Left: data for $n = 10000$. Right: error rates of the two versions of lookout, with points jittered horizontally.*

they move from the inner edge of the annulus towards the centre of the hole. Each iteration is repeated 10 times. Fig. 5 shows the data at iterations 3 and 9 and the results. The AUC and Gmean of the new version are close to one from the seventh iteration, when it identifies all five anomalies in every replication; its Fmeasure levels off at about 0.77 because of false positives. The original version is lower and more variable, KDEOS and RDOS have AUC of at most about 0.5 throughout, and HDoutliers and stray identify almost no anomalies before the eighth and tenth iterations respectively.

4.5 Experiment 5: Unit cube in $\mathbb{R}^{20}$

The sample has 500 points in $\mathbb{R}^{20}$, of which one is an anomaly. Initially all 500 points are uniform on $(0, 1)$ in each dimension. At iteration $i \in \{1, \dots, 20\}$ the first i coordinates of the anomaly are

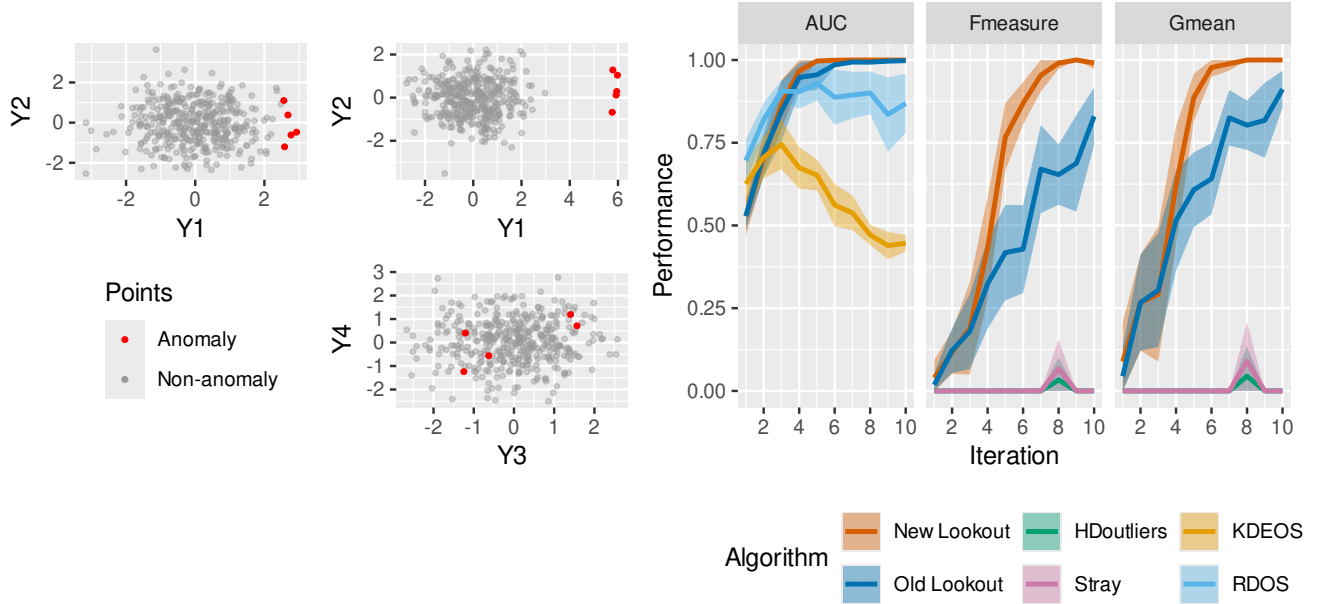


Figure 4: *Experiment 3. Left: the Y_1Y_2 plane at the third iteration, with anomalies in red. Middle: the Y_1Y_2 (top) and Y_3Y_4 (bottom) planes at the ninth iteration. Right: mean AUC, Fmeasure and Gmean over ten replications, with bands of ± 2 standard errors, for the two versions of lookout, HDoutliers, KDEOS, RDOS and stray.*

set to 0.9, so that at iteration 20 it is $(0.9, \dots, 0.9)$, and each iteration is repeated 10 times. The anomaly is not anomalous in any single coordinate, on any $Y_i Y_j$ plane or in the first two principal components, so Fig. 6 shows the $D_1 D_2$ plane of the projection given by *dobin* (Kandanaarachchi & Hyndman 2021), a dimension reduction method designed for anomalies, at iterations 5, 12 and 20, together with the results. The anomaly is mixed with the other points at iteration 5 and separated from them at iterations 12 and 20. Both versions of lookout are run without scaling. The AUC of the new version is close to one from the twelfth iteration except at the fourteenth, and that of RDOS from about the eleventh; the original version and KDEOS are lower. The Gmean of the new version is close to one from the sixteenth iteration, when it identifies the anomaly in every replication. Its Fmeasure stays below 0.5 because of about three false positives among the 499 non-anomalies. HDoutliers and stray never identify the anomaly.

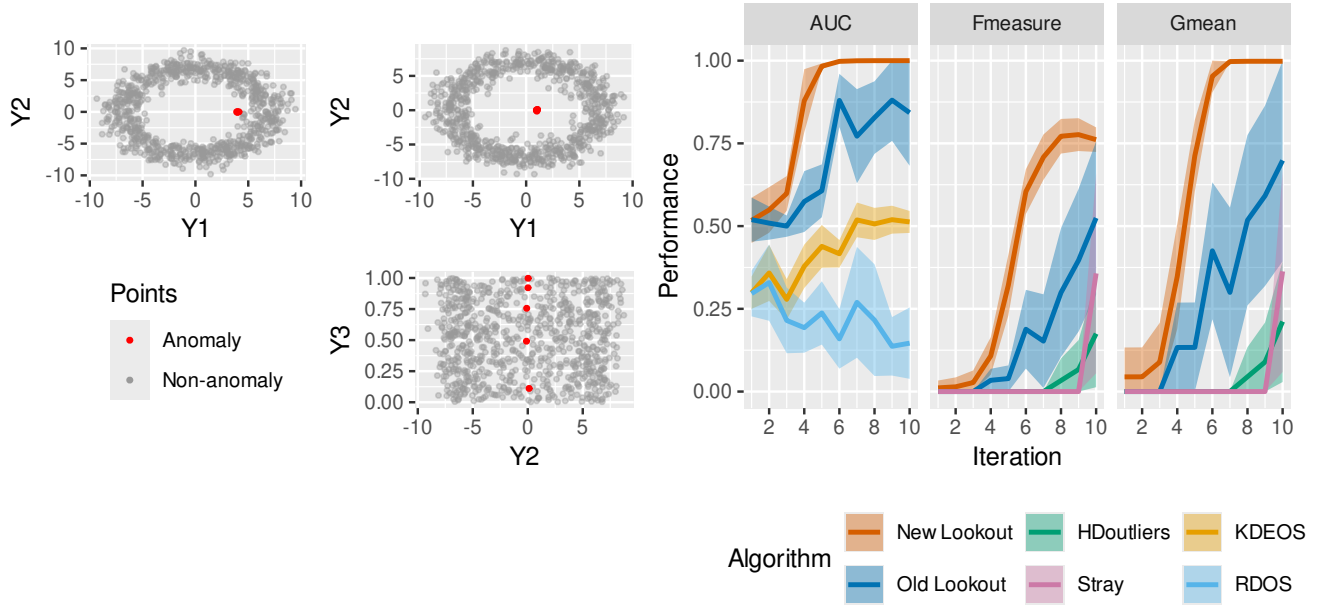


Figure 5: *Experiment 4. Left: the $Y_1 Y_2$ plane at the third iteration, with anomalies in red. Middle: the $Y_1 Y_2$ (top) and $Y_2 Y_3$ (bottom) planes at the ninth iteration. Right: mean AUC, Fmeasure and Gmean over ten replications, with bands of ± 2 standard errors, for the two versions of lookout, HDoutliers, KDEOS, RDOS and stray.*

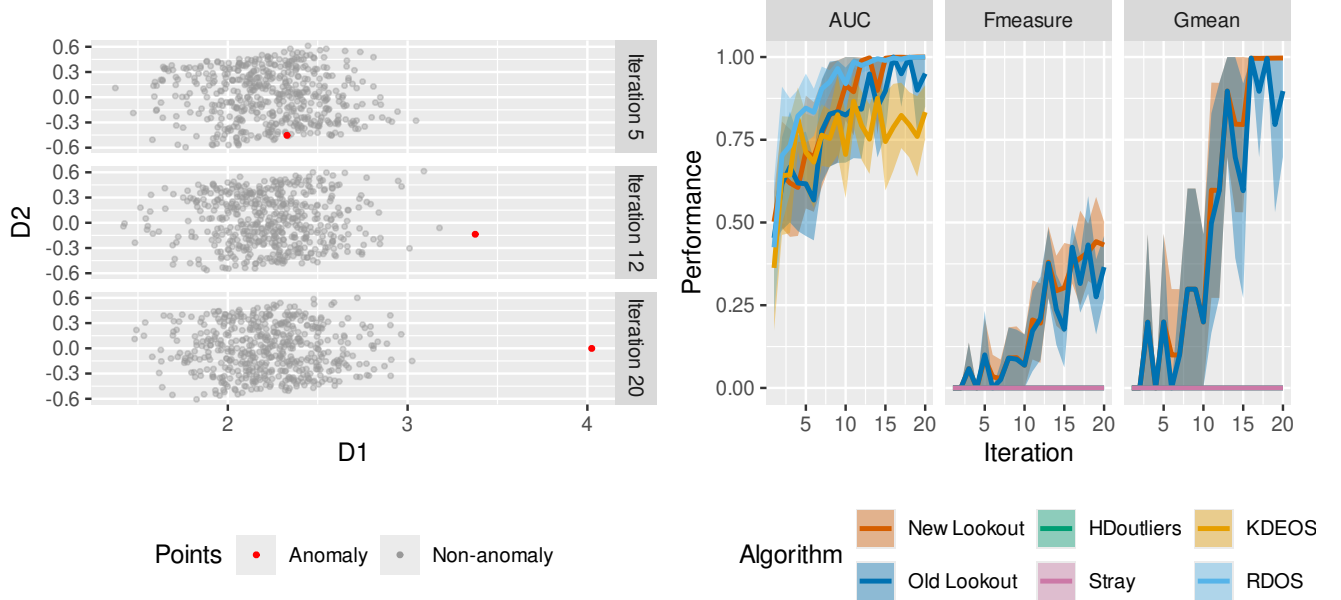


Figure 6: *Experiment 5. Left: the D_1D_2 plane of the dobin projection at the fifth, twelfth and twentieth iterations, with the anomaly in red. Right: mean AUC, Fmeasure and Gmean over ten replications, with bands of ± 2 standard errors, for the two versions of lookout, HDoutliers, KDEOS, RDOS and stray.*

4.6 Sensitivity to the bandwidth

Section 2.3 argues that the bandwidth should be of the order of the spacing in the tails, and that neither the constant nor MISE-optimality matters much. To check this, we repeated Experiments 1, 2 and 3, and Experiment S1 of the online Appendix, with h_n multiplied by 1/4, 1/2, 2 and 4, and with h_n replaced by the AMISE-optimal bandwidth h_{opt} of Table 1. The details and results are in the online Appendix. The area under the ROC curve was almost the same for every bandwidth from $h_n/2$ to $4h_n$, apart from three iterations of Experiment 3 at $4h_n$, so the ranking of the observations is insensitive to the bandwidth. The threshold is more sensitive. At h_n the false positive rate was close to the nominal $\alpha = 0.01$, and at $h_n/4$ it was between 0.05 and 0.27, because observations with no other observation inside the kernel support are all declared anomalous. With

h_{opt} the results were similar to those at h_n for normal data in two dimensions. In six dimensions h_{opt} was about a quarter of h_n .

5 Applications

We apply the new and original versions of lookout to two real datasets. Neither has labelled anomalies, so accuracy cannot be measured, but both have two variables, so the output can be inspected.

1. Old Faithful dataset¹: 2,097 eruptions of the Old Faithful Geyser in Yellowstone National Park, Wyoming, USA, from 14 January 2017 to 29 December 2023. For each recorded eruption the duration is given in seconds along with the waiting time to the following eruption. The records are incomplete, especially during winter.
2. Wine reviews dataset²: 110,203 reviews of wines from 42 countries, obtained from the *Wine Enthusiast Magazine* during the week of 15 June 2017. We use the 4,496 wines labelled Shiraz or Syrah, and two variables: the points score given by the *Wine Enthusiast* reviewer on a scale of 0–100, and the price of a bottle in US dollars. The kernel density estimates use the truncated sum described in Section 3, with 100 neighbours.

Fig. 7 shows the Old Faithful data. The new version flags 25 eruptions in the sparse regions between and around the two clusters. The original flags 15, nine of which are also flagged by the new version; the other six form a group at the lower edge of the smaller cluster.

Fig. 8 shows the wine reviews. The new version flags 66 wines and the original 67, with 42 in common. Both flag the most expensive wines at scores from 89 to 96. The new version also flags the most expensive wine at six of the eight scores from 81 to 88, although these prices are moderate: at 85 points it flags two wines at \$80, where the median price is \$21. Of the 52 wines

¹Data downloaded from <https://geysertimes.org>.

²Data downloaded from <https://www.kaggle.com/datasets/zynicide/wine-reviews/data>.

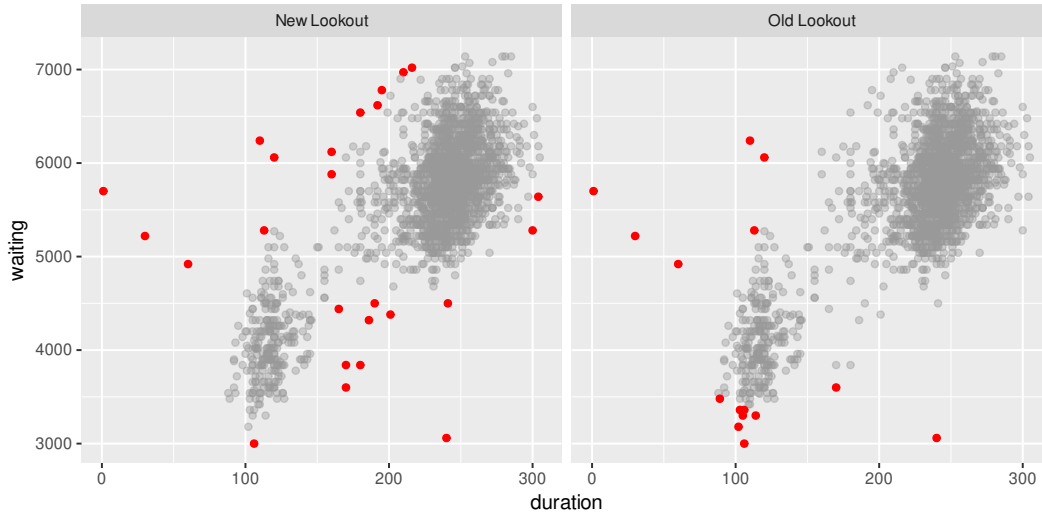


Figure 7: *Anomalies (red) identified in the Old Faithful data by the new version of lookout (left) and the original version (right).*

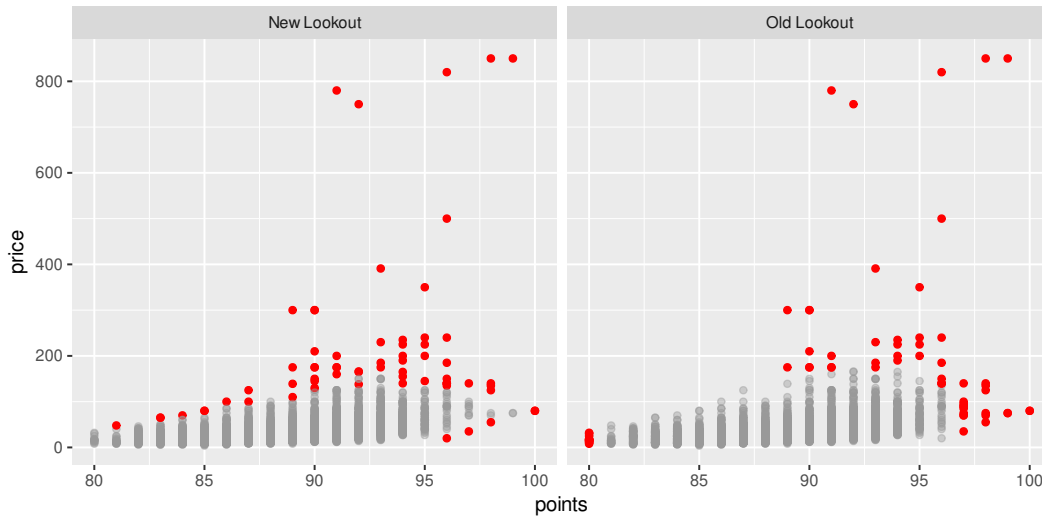


Figure 8: *Anomalies (red) identified in the wine reviews data by the new version of lookout (left) and the original version (right).*

scoring above 95, it flags 20, including the cheapest wines at 96, 97 and 98 points. The original flags no wine scoring 81 to 88. It flags all 13 wines with the lowest score of 80 and all 22 wines scoring 97 or more, whatever their price. Most of this difference comes from the scaling. After min-max scaling one point of score is equivalent to \$42, and after minimum covariance determinant

(MCD) scaling to about \$5, so the original treats a one-point gap in score as large, and the new version treats it as small compared with gaps in price.

In both datasets the new version flags observations in sparse regions of the two-dimensional space. The original also flags observations at the edge of the data that are close to other observations: the lower edge of the smaller Old Faithful cluster, and the wines with the lowest and highest scores. We regard the behaviour of the new version as the more useful.

6 Discussion and conclusions

Working with the minimum spanning tree in place of the Rips filtration has allowed us to analyse the bandwidth of the lookout algorithm. The quantile bandwidth h_n has the order $n^{-1/m}$ of the spacing between neighbouring observations for any density, while the largest-gap bandwidth of the original algorithm can grow without bound under heavy tails. The resulting density estimate is not MISE-consistent. In [Section 4.6](#) the ranking of the observations changed little when h_n was multiplied by factors from 1/2 to 4. In the simulations the new version performs better than the original or about as well, except at the largest sample sizes of Experiment S2. It is ahead of the four other methods, except that RDOS has a higher AUC in the early iterations of Experiments 3 and 5.

Two choices affect which observations are identified. The first is the quantile level. By the single linkage interpretation of [\(7\)](#), at most a proportion of about $1 - \gamma$ of the observations are isolated at h_n . If a larger proportion are isolated anomalies, h_n is set by the spacing between them, so a smaller γ should be used when a large proportion of anomalies is expected. The second is the scaling. The new version standardizes the data with the full MCD covariance matrix, so the tree and the kernel density estimate depend on the data only through robust Mahalanobis distances between observations. Since the MCD estimator is affine equivariant, the observations identified are unchanged by any invertible affine transformation of the data. Min-max scaling is unaffected only by shifting and rescaling each variable separately, and it accounts for most of the

difference between the two versions in the wine reviews data. The MCD estimate requires more observations than variables, and is unstable when n is less than about $2m$. When its covariance matrix is singular, the implementation adds a small ridge to it. The scaling can also be turned off, as in Experiment 5, where all variables have the same scale.

The algorithm applies to any continuous multivariate data, and in the experiments it performs well in up to 20 dimensions. It can be applied to problems that can be transformed to that setting, such as image anomaly detection through feature extraction or functional data through functional principal components. It assumes independent and identically distributed observations, so temporal or spatial data need pre-processing, for example by applying the algorithm to the residuals of a fitted model. In high dimensions kernel density estimation suffers from the curse of dimensionality, and dimension reduction such as principal components or dobin ([Kandanaarachchi 2022](#)) before applying lookout may help.

The tree is computed in $O(n \log n)$ time ([March et al. 2010](#)), and the cost is dominated by the kernel density estimate, for which the implementation uses a truncated sum over nearest neighbours when $n > 10^4$ ([Section 3](#)).

Two theoretical questions remain open. We have no formula for where the largest gap between consecutive edge lengths occurs, so the bandwidth of the original algorithm has not been analysed beyond the results on the longest edge. The constant $q_\gamma(f)$ of [Proposition 1](#) is defined through the component count of a Poisson process and is not explicit, although its bounds t_- and t_+ are.

Supplementary material

The online Appendix contains the proofs of [Proposition 1](#) and [Counterexample 1](#), two further comparisons of the original and modified algorithms, a set of showcase examples, and the details and results of the sensitivity study of [Section 4.6](#). The updated algorithm is implemented in the `lookout` R package ([Kandanaarachchi et al. 2026](#)), available on CRAN at <https://CRAN.R>

[-project.org/package=lookout](https://project.org/package=lookout). Files to reproduce all numerical examples are available at <https://github.com/robjhyndman/lookout2>.

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Disclosure statement

The authors report there are no competing interests to declare.

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